

Abstract

This work presents a supervised text classification approach for emotion detection in social media text. A machine learning pipeline consisting of text preprocessing, TF-IDF vectorization, and Logistic Regression classification was implemented and evaluated using an emotion-annotated Twitter dataset obtained from Kaggle. The dataset contains six emotion categories: sadness, joy, love, anger, fear, and surprise. Experimental results demonstrate that the proposed model achieves strong overall performance, with high precision, recall, and F1-scores across most emotion classes. Performance variations across classes highlight the impact of dataset imbalance, particularly for minority emotions. These results indicate that lexical features captured through TF-IDF representations are effective for emotion classification in short informal text.

Introduction

Text classification is a machine learning task that involves assigning predefined labels to textual data in order to automatically categorize it into meaningful groups (Stryker, n.d.). Text classification techniques can be applied in either supervised or unsupervised settings, with supervised approaches relying on labeled data to learn relationships between text and target categories.

One important application of text classification is emotion detection, which aims to identify specific emotional states expressed in text rather than simple positive or negative sentiment. Emotion classification is particularly challenging in social media data, where texts are short, informal, and often contain slang or non-standard language. These characteristics make it difficult to reliably capture emotional cues using traditional linguistic features.

In this work, a supervised text classification model is developed to classify emotions in Twitter messages using an emotion-annotated dataset. The proposed approach employs text preprocessing, TF-IDF feature representation, and a Logistic Regression classifier to model lexical patterns associated with different emotional categories.

Dataset & Preprocessing

The emotion classification dataset provided by Kaggle was used in this study. It consists of English-language tweets annotated with six emotion labels encoded as numerical values: sadness (0), joy (1), love (2), anger (3), fear (4), and surprise (5).

The dataset was split into training and test sets using an 80/20 ratio with stratified sampling to preserve label distribution. Text preprocessing was applied after splitting to avoid information leakage and included lowercasing, punctuation and non-alphabetic character removal, and stopword removal to reduce noise while retaining meaningful lexical content.

Feature Analysis

An analysis of the training data was conducted to examine key linguistic characteristics relevant to emotion classification. The distribution of emotion labels is highly imbalanced, with *joy* being

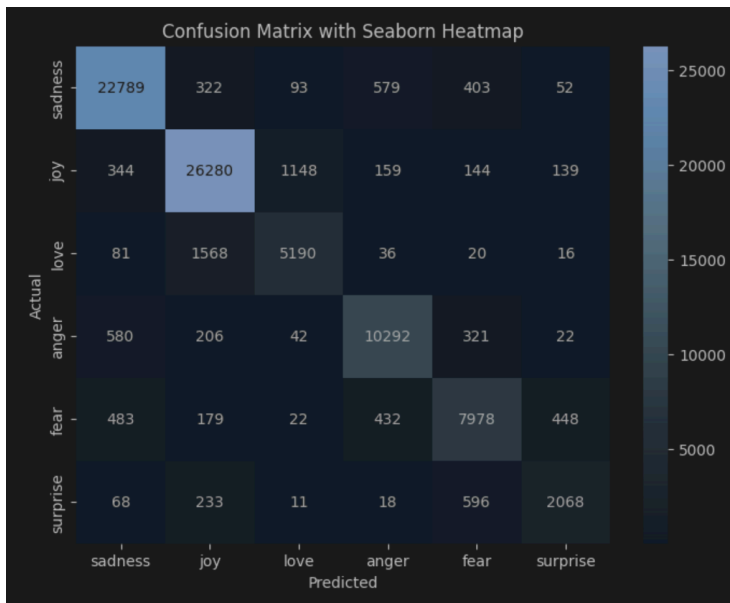
the most frequent class and *surprise* the least frequent, indicating a potential bias toward majority classes during classification.

Tweet length analysis shows that texts are generally short, with an average length of approximately nine words. Average text length remains consistent across emotion categories, suggesting that emotional distinctions are primarily conveyed through lexical content rather than structural differences.

Lexical analysis of the training data reveals frequent use of first-person and affective terms such as “*feel*”, highlighting the importance of word-level features in capturing emotional expression. Overall, these findings indicate that short informal text and emotion-specific vocabulary are key linguistic characteristics of the dataset, motivating the use of TF-IDF based supervised classification models.

Model & Evaluation

	Precision	Recall	f1-score
0	0.94	0.94	0.94
1	0.91	0.93	0.92
2	0.80	0.75	0.77
3	0.89	0.90	0.90
4	0.84	0.84	0.84
5	0.75	0.69	0.72
Accuracy	0.89		



The Logistic Regression model was evaluated on the test set using precision, recall, F1-score, and accuracy. The classifier achieved an overall accuracy of 89%, indicating strong performance on the emotion classification task.

Performance is highest for majority classes such as *sadness* and *joy*, while minority classes such as *love* and *surprise* show lower recall and F1-scores. This reflects the imbalanced nature of the dataset and

highlights the importance of considering macro-averaged metrics in addition to accuracy. The macro-averaged F1-score of 0.85 indicates reduced performance on less frequent classes, whereas the weighted F1-score of 0.89 demonstrates strong overall performance.

The confusion matrix shows that most predictions fall along the diagonal, indicating correct classification across emotions. Misclassifications primarily occur between semantically related emotions such as *joy* and *love*, suggesting overlapping lexical patterns. Overall, the results demonstrate that TF-IDF-based Logistic Regression is effective for emotion classification in short social media text.

Conclusion

This work presented a supervised text classification approach for emotion detection in social media text using TF-IDF feature representation and Logistic Regression. The model achieved strong overall performance, with an accuracy of 89% and high precision and recall across most emotion classes. Performance differences across classes highlight the impact of dataset imbalance, particularly for minority emotions. Future work could explore alternative feature representations or class balancing techniques to further improve performance on less frequent emotion categories.

Reference

Cole Stryker (n.d.) What is Text Classification? IBM. Available at:
<https://www.ibm.com/think/topics/text-classification>